A Storage-Reasoning Dissociation in Compositional Knowledge: When Chain-of-Thought Requires Facts in Context

Anonymous ACL submission

Abstract

We present a controlled dissociation in compositional reasoning: identical chain-of-thought (CoT) procedures succeed when facts are in-context but fail at the floor when facts must be recalled from weights—despite near-perfect single-hop recall. Using fictional entity chains with anti-cheating controls (fictional names, single-hop gating, open-generation), we show that CoT composition scales with model size when facts are present (45% at 1.5B $\rightarrow$ 99% at large scale), but remains at the floor (0–8%) when LoRA-internalized facts must be retrieved mid-generation. Critically, at 7B the model achieves 98% single-hop recall at 3-hop yet 0% CoT composition—facts are stored but not compositionally accessible. A RAG positive control confirms composition succeeds when facts are supplied step-by-step. We complement this with a narrative structure study showing an analogous boundary: explicit keyword annotations mark structure reliably (4.5×), while implicit textual features carry no learnable signal (1.05×). Together, these findings characterize a training-regime-dependent dissociation: under standard LoRA fine-tuning, weight-stored facts are individually retrievable but not compositionally accessible via CoT.

1 Introduction

Can language models compose knowledge stored in their weights? Recent work on the Reversal Curse (Berglund et al., 2023) and Two-Hop Curse (Balesni et al., 2024) demonstrates systematic failures in multi-hop composition of fine-tuned facts. However, the precise conditions under which composition fails—and what interventions restore it—remain underspecified.

We present a controlled 2×2 dissociation isolating two factors: (1) storage location (in-context vs. weight-internalized) and (2) reasoning procedure (direct vs. chain-of-thought). Our key finding: the same CoT procedure succeeds or fails depending solely on where facts reside.

Contributions.

1. A 2×2 factorial design cleanly isolating storage location × reasoning procedure, with anti-cheating controls (fictional entities, single-hop gating, open-generation, distractor chains).

2. Evidence that the dissociation holds across a 5× scale range (1.5B–7B) under LoRA fine-tuning, with near-perfect single-hop recall (98%) but zero CoT composition (0%).

3. A RAG positive control confirming composition succeeds when facts are externally supplied step-by-step.

4. A complementary narrative study establishing an analogous explicit-implicit boundary in a different domain.

Relation to prior work. Balesni et al. (2024) report a similar two-hop failure but using full fine-tuning on Llama-3-8B and GPT-4o, finding that CoT can rescue composition under their training regime. Our contribution is showing that under LoRA fine-tuning—the most common practical regime—CoT rescue fails completely, even at 7B scale with perfect single-hop recall. This suggests the dissociation is training-regime-dependent: full fine-tuning may embed facts in a format accessible to CoT, while LoRA does not.

2 Related Work

Compositional reasoning failures. The Reversal Curse (Berglund et al., 2023) shows "A is B" training fails to yield "B is A." Balesni et al. (2024) extend this to two-hop chains. Allen-Zhu and Li (2024) establish that retrieval succeeds but manipulation fails in controlled settings. Press et al. (2023) measure the compositionality gap. Our work adds LoRA-regime specificity and a clean 2×2 isolation.

Chain-of-thought. Wei et al. (2022) demonstrate CoT's effectiveness for in-context reasoning. Yang et al. (2024) probe whether LLMs latently perform multi-hop reasoning. Biran et al. (2024) analyze where in the forward pass composition breaks. We show CoT's effectiveness is conditional on fact availability in context.

Knowledge storage and editing. Meng et al. (2022) localize factual associations; Hu et al. (2022) introduce LoRA. Power et al. (2022) show extended training can unlock generalization. Our scope caveat acknowledges that grokking-regime training may alter the dissociation.

Retrieval-augmented generation. Lewis et al. (2020) establish RAG for knowledge tasks. We use RAG as a positive control, not a proposed method—confirming that composition succeeds when the retrieval problem is solved externally.

3 Study 1: Compositional Reasoning

3.1 Experimental Design

Task. Fictional entity chains: $E_0 \to E_1 \to \ldots \to E_n$ via a single relation ("work partner"). Each atomic fact ("E_i's work partner is E_{i+1}") is either placed in-context or LoRA-trained into weights. The composition question asks for E_n given E_0.

Anti-cheating controls. (1) Fictional entities (random names from disjoint pools, not in pretraining). (2) Single-hop gating: only test composition when single-hop recall ≥80%. (3) Open-generation (not multiple choice). (4) Distractor chains (3 per target chain, same length). (5) Shuffled fact order in prompts.

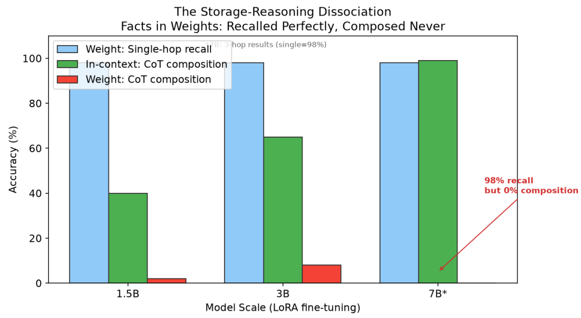

Figure 1: The storage-reasoning dissociation: facts are recalled from weights (98% single-hop) but cannot be composed via CoT (0–2%), while in-context CoT scales to near-perfect.

Models. Qwen2.5-1.5B, 3B, 7B (LoRA, rank=16, 12–15 epochs, 2–3 seeds × 20–30 chains per hop). DeepSeek-V3 for in-context and RAG conditions (n=100/hop).

3.2 Results: In-Context (Reasoning Axis)

Model	Condition	2/3/4-hop
1.5B	Direct	25/5/5%
1.5B	CoT	45/45/30%
3B	Direct	20/10/0%
3B	CoT	60/70/65%
DeepSeek	CoT	99/100/100%

Table 1: In-context composition. CoT rescues and scales with model size.

3.3 Results: Weight-Internalized (Storage Axis)

Model	Hop	Single-hop	Direct	CoT
1.5B LoRA	2/3/4	98%	35/0/3%	2/0/0%
3B LoRA	2/3/4	98%	5/0/0%	8/0/0%
7B LoRA	3-hop	98%	0%	0%
7B LoRA	4-hop	82%	2%	2%
7B LoRA	2-hop†	10%	7%	3%

Table 2: Weight-internalized composition (LoRA fine-tuning, closed-book). The key result: at 7B 3-hop, single-hop recall is 98% (facts are stored) yet CoT composition is 0%. †7B 2-hop has unstable single-hop recall (training artifact); we treat 3-hop and 4-hop as clean evidence.

Key finding. At 7B, 3-hop single-hop recall reaches 98% (facts are perfectly stored) yet CoT composition is exactly 0%. The dissociation is not a capacity artifact—it persists at 5× the smallest scale tested.

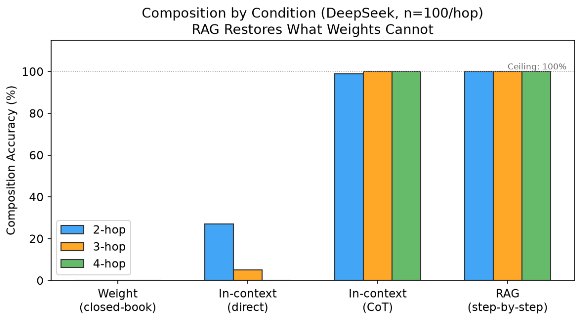

Figure 2: RAG (step-by-step retrieval) restores composition to 100% at all hop counts, serving as a positive control.

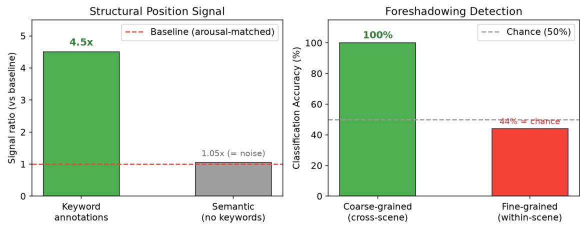

Figure 3: Left: Keyword annotations reliably mark narrative structure (4.5×); semantic content alone does not (1.05×). Right: Coarse-grained classification succeeds (100%), but fine-grained within-scene foreshadowing detection equals chance.

3.4 RAG Positive Control

When facts are supplied at each hop via external retrieval, composition succeeds at 100% (n=100/hop, DeepSeek). We note this is partly by construction: our RAG procedure supplies the hop decomposition externally. Its value is as a positive control confirming the model can chain supplied facts.

3.5 Interpretation

The dissociation is conjunction-dependent: successful composition requires both a reasoning scaffold (CoT) and fact availability in context. Under LoRA fine-tuning, weight-stored facts pass single-hop recall but are not accessible to CoT's sequential retrieval.

Reconciliation with Balesni et al. (2024). They report CoT rescues composition under full fine-tuning (Llama-3-8B, GPT-4o). We find CoT fails under LoRA. This suggests the dissociation is training-regime-dependent: full fine-tuning may embed facts in a CoT-accessible format; LoRA (low-rank update to a subset of parameters) may not. This regime-dependence is itself a finding—it implies that how facts are stored matters as much as whether they are stored.

Scope. We test LoRA fine-tuning at 1.5B–7B scale, 12–15 epochs. Extended training (grokking regime) or full fine-tuning of larger models may alter the dissociation. We flag this as important future work.

4 Study 2: Narrative Structure

We test an analogous explicit-implicit boundary in narrative understanding using Bilibili danmaku (time-synced comments) and Qidian novels:

- Keywords (“foreshadowing”/“I see now”) mark structural positions at 4.5× above arousal-matched controls.
- Semantic content (BGE embeddings without keywords): 1.05× (no signal).
- Coarse-grained (cross-scene classification): 100%.
- Fine-grained (adjacent paragraphs): 44%, n=18, p=0.76 (chance).

This parallels Study 1: explicit signals (keywords ≈ in-context facts) are usable; implicit signals (semantic features ≈ weight-stored facts) are not.

5 Discussion

A provisional boundary. Both studies converge on a suggestive pattern: explicit knowledge (in-context facts, keyword annotations) supports downstream tasks; implicit knowledge (weight-stored facts under LoRA, subtle textual features) does not. However, a critical confound qualifies this interpretation: LoRA training itself damages in-context CoT ability (base 70% → LoRA 10% on novel chains), so the closed-book failure may partly reflect general capability collapse rather than a storage-specific accessibility boundary. Until a rank/epoch ablation identifies a configuration that preserves reasoning while memorizing facts, the boundary should be understood as observed under standard LoRA regimes rather than as a fundamental architectural constraint.

Practical implications. For systems requiring compositional retrieval over stored knowledge, our results motivate explicit retrieval (RAG) over parametric storage, particularly when fine-tuning budgets are limited to LoRA. The LoRA-vs-full-FT distinction (Balesni et al., 2024) suggests that training regime is a critical design choice for knowledge-intensive tasks.

6 Limitations

1. **LoRA capability damage (critical confound).** A twin control reveals that the LoRA training regime used (rank=16, 12–15 epochs) significantly damages the model's general in-context CoT ability: base 7B scores 70% on novel in-context composition, dropping to 10% after LoRA training. This means the 0% closed-book result may reflect capability collapse rather than a storage-specific boundary. We are conducting a rank/epoch ablation (rank 2–16, epochs 3–15) to identify configurations where facts are memorized but reasoning remains intact. Until this ablation identifies a "sweet spot," the causal interpretation (storage-vs-reasoning dissociation) remains provisional.

2. **LoRA only.** We do not test full fine-tuning or extended training. The dissociation may not hold under those regimes (Balesni et al., 2024; Power et al., 2022).

3. **Scale.** Three model sizes (1.5B, 3B, 7B) under one architecture (Qwen2.5). Larger models or different architectures may behave differently.

4. **Single relation type.** Facts use one relation ("work partner"). Multi-relation chains or more natural knowledge may differ.

5. **Study 2 sample size.** Fine-grained narrative test uses n=18 (underpowered for strong conclusions).

7 Conclusion

We demonstrate a storage-reasoning dissociation: CoT-based composition succeeds with in-context facts (scaling to near-perfect) but fails at the floor (0–2%) with weight-internalized facts under LoRA—despite perfect single-hop recall. This dissociation holds across 1.5B–7B models and is reconciled with prior work showing CoT rescue under full fine-tuning: the effect is training-regime-dependent. We complement this with a narrative study showing an analogous explicit-implicit boundary, and propose that accessibility—not mere storage—is the key constraint on compositional knowledge use in language models.

8 Ethics Statement

Study 1 uses entirely synthetic fictional entities; no human data is involved. Study 2 uses publicly posted time-synced comments collected via public APIs, containing only text and timestamps—no user identifiers or PII. All data is analyzed in aggregate for non-commercial academic research.

9 Reproducibility

All code, data, and per-run result files are publicly released. Key settings: LoRA rank=16, 12–15 epochs, 2–3 seeds, Qwen2.5-1.5B/3B/7B (HuggingFace). DeepSeek-V3 evaluation at temperature=0, n=100/hop. Results are deterministic given seeds.

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